view of that. Basically the same scene, instead of being viewed in 2D from one angle of the camera you can now see it from different directions. So the person catching the ball or getting out, you can see it from multiple directions and you wonder how that happens. Typically what happens is there's a whole bunch of cameras in the stadium and all those cameras are recording this scene simultaneously and it is sent to the studio. In the studio, people put all the videos together and create a 360-degree view of the action. Then a person such as a producer, will pick a particular path of the viewpoint through the 360 degree space. And then you get this view of the camera rotating around the field.

The problem still is that there's only one 360-degree view. Now imagine a scenario where all these videos are shipped simultaneously to your phone because you probably have the bandwidth by that time. And then you can choose the view that you want to see. That's where entertainment is going.

When you do that, you've got a bunch of 4K videos on your phone. You have to first decode them, next you have to stitch them, then you have to create the 360-video and then re-encode the video that can be watchable. And then finally create that and play. Imagine the amount of memory and processing needed to make that happen. That's what people want.

There was a time when live video could not be paused but now, no one even thinks about it. It always surprises me pleasantly, the applications that come when you create the infrastructure on the client and the cloud, followed by the operating system and the products that can work with all of the above. In my 25-30 years, I've seen a couple of things get overrun. First is bandwidth, second is computation power and third is memory and storage. We've never been able to put enough of bandwidth or computation power or memory.

Even though the Micron-MediaTek LPDDR5 validation happened just now, you'd mentioned that the products would arrive in the latter half of 2022. Could you take us through the process?

Raj: What happens at this stage is that we typically provide samples to all of our customers. And then MediaTek will provide their products to the customers. And our customers are busy making the products. We don't control the schedule at that point.

Would you term LPDDR5X as a drop in replacement for LPDDR4X or are there some significant design changes required?

Raj: We need a new processor that supports LPDDR5X. We can't do a drop-in with a processor that doesn't support it. We need to work with a processor that has the LPDDR5X interface and supports the performance capabilities.

Would there be a cost increase for your customers upgrading their products from LPDDR4 to LPDDR5X? Let's say in terms of the bill of materials cost for their latest smartphone.

Raj: Clearly, new memory costs more but the performance is higher. So they might be able to translate that into a higher performance phone and maybe they might be able to get higher margins for that product. The use cases that LPDDR5X will support are numerous. They could have more cameras, better speed, more AI applications.

When you have a higher-bandwidth memory, and if the processor is also able to run at a higher speed, then the processor and memory will be able to finish the workload sooner. That can translate into better battery life. It's not so intuitive but it's actually something that happens. Let's say that you need to do some amount of work. You can run at a certain clock speed for some amount of time and get the workload time. Or you could run at a higher clock speed for a shorter period of time and get the work done quicker. The amount of time required for the workload is lesser so the amount of battery consumed could also be lesser. That's another benefit that higher bandwidth memory brings to the table.

On the desktop side of things, we just saw the transition from DDR4 to DDR5. One thing that came to light was that because of the shortage of PMIC (Power Management IC) chips, RAM manufacturers couldn't build enough DIMMs. And that's leading to a shortage. On the mobile front, do you need an external PMIC or can the SoC handle the power management?

Raj: It's the SoC PMIC. You don't need a separate PMIC.

What about supply? Is that going to be an issue with LPDDR5X?

Raj: No, we're ready to ship. It's on our same 1α Node and we've invested a lot of CAPEX on that. So we're able to deliver.

In terms of the AI applications, fast memory usually helps in applications that use large frames of data such as video or image processing. Aside from those, what other AI applications do you feel will benefit from faster memory?

Raj: I've built camera processors for 25 years. The most amount of AI used on phones is in the camera. Let's say that you're trying to take a picture at night. Now there's a feature called Night Mode. What Night Mode basically is, it takes a bunch of pictures one after the other really fast, registers them, aligns them onto each other, figures out what part of one picture is the same as another, and then integrates all of them to create one good image. And meanwhile, if your hands shake, it figures out how to take the shake out of the image. So it's performing motion compensation, handshake reduction, combining the pictures, stitching them together and it's using a tonne of AI for all this.

The other major challenge for cameras is called the 3A problem. It stands for auto-exposure, auto-whitebalance and auto-focus. This is something that humans are able to do really well. They're able to focus (foveate), they're able to figure out the white balance and they're able to open their pupils wider (auto exposure). What AI tries to do is take a whole bunch of images and see what the right exposure, focus and white balance is for the scene. So that's why the camera is the #1 use case for AI. And that's what needs the most amount of bandwidth. That's why I believe that LPDDR5 and LPDDR5X are going to continue to sell, because the #1 use case needs it.